

BHANU KIRAN R

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SUMMARY

Software Development Engineer specialising in scalable, production-grade systems with expertise in full-stack development (React.js, React Native, FastAPI, Flask) and AI/ML integration (Large Language Models, Retrieval-Augmented Generation, Natural Language Processing). Proven ability to architect high-concurrency applications handling 200+ concurrent connections with sub-500ms latency. Published researcher in AI-driven pharmaceutical applications. Technical leader demonstrating cross-functional collaboration, project management, and stakeholder engagement while mentoring developers in software engineering best practices, including code reviews, design patterns, and CI/CD pipelines.

EDUCATION

C Byregowda Institute of Technology

Kolar, Karnataka

Bachelor of Engineering in Computer Science; CGPI: 8.1

2023 – 2027

- Member of Club of Programmers (COPS CBIT) - Direct technical initiatives for 200+ student community
- Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Database Systems

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, HTML/CSS, LaTeX

AI/ML: TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, LangChain, OpenAI API, Groq LLM, RAG

Backend: FastAPI, Flask, Node.js, RESTful APIs, gRPC, WebSockets, Microservices, Flask-SocketIO

Frontend: React.js, React Native, Expo, Tailwind CSS, Bootstrap, Mapbox GL JS, Jinja2

Databases: PostgreSQL, MySQL, MongoDB, NoSQL, PostGIS

DevOps & Tools: Docker, Kubernetes, Git, GitHub Actions, CI/CD, Linux, Render

Core CS: Data Structures & Algorithms, System Design, Object-Oriented Design, Design Patterns, Concurrent Programming

PROJECTS

IPC Legal Assistant - Cross-Platform Mobile Application | React Native, Flask, LangChain, Groq LLM, RAG Jan 2025

- Developed a full-stack cross-platform mobile application providing instant, verified information on the Indian Penal Code 1860 using React Native (Expo) and Python Flask backend with cloud deployment on Render.
- Implemented Retrieval-Augmented Generation (RAG) architecture using LangChain and Llama 3.3-70B to provide citations from a 227-page legal corpus, achieving 92% accuracy in legal section retrieval.
- Engineered a hybrid keyword-matching and semantic search system to ensure high precision in legal section retrieval, reducing query response time to under 800ms through optimised vector similarity algorithms.
- Orchestrated a complete CI/CD deployment pipeline using GitHub and Render, optimising the backend to run efficiently on 512MB RAM with 99.5% uptime over a 60-day production testing period.

VIKRANTA - Tourist Guardian Platform | React, Flask, PostgreSQL, PostGIS, Docker Oct 2025

- Engineered a real-time safety system achieving a 98% alert delivery rate by architecting a WebSocket-based communication layer handling 50+ concurrent distress signals with sub-3s notification latency.
- Reduced geospatial query response time by 85% (under 100ms) by implementing PostGIS spatial indexing on 10,000+ danger zones for real-time safety score calculations.
- Built an AI-driven cultural guide using Retrieval-Augmented Generation (RAG) with Gemini AI, improving multilingual content retrieval speed by 60% across 15+ location parameters.
- Deployed containerised microservices on Railway with horizontal auto-scaling, supporting 200+ active tourist sessions during peak load testing.

Python AI Chatbot | FastAPI, OpenAI API, WebSockets, Redis, Docker Dec 2024

- Developed an asynchronous FastAPI platform handling 200+ concurrent WebSocket connections with 99.8% uptime and zero message loss over a 90-day production period.

- Reduced average LLM response latency to 450ms by optimising OpenAI API integration and implementing Redis session management for efficient context pooling.
- Decreased hallucination rates by 40% through advanced prompt engineering and context window management supporting 15+ message exchanges.
- Orchestrated an automated CI/CD pipeline using GitHub Actions, reducing release cycle time by 75% for containerised deployments.

ImageClassifier - Deep Learning Vision | TensorFlow, Keras, ResNet50, Python

Nov 2025

- Achieved 94.3% top-1 accuracy on 50,000+ images by architecting a multi-class CNN with optimised layer configurations and adaptive learning rate scheduling.
- Reduced training time by 60% by utilising transfer learning with a pre-trained ResNet50 backbone and fine-tuning specific residual blocks.
- Applied TensorFlow Lite quantisation to reduce model size by 75% (98MB to 24MB), decreasing CPU inference latency from 145ms to 38ms for edge readiness.

PUBLICATIONS & RESEARCH

AI Applications in Pharmaceutical Drug Discovery

NCRTEST-2025

- Published research exploring Graph Neural Networks (GNNs) for molecular property prediction, analyzing drug-target binding affinity across 5,000+ compound structures.
- Presented at 7th National Conference on Recent Trends in Engineering Science and Technology (NCRTEST-2025), contributing methodology for accelerating early-stage drug screening.
- Investigated deep learning approaches, reducing computational requirements for compound analysis by 35% compared to traditional molecular dynamics simulations.

ACHIEVEMENTS & CERTIFICATIONS

Certificate of Appreciation - Hack Fest'25

Bangalore Technological Institute, Apr 2025

- Recognized among the top 5 teams from 40+ competing groups for innovative solution to a real-world technical challenge.

AI Unplugged: Machine Learning, Deep Learning & Generative AI

DHS Informatics, Oct 2025

- Completed 40-hour intensive training covering supervised/unsupervised learning, neural network architectures, and LLM applications.

Hands-on Computer Networks using Cisco Packet Tracer

Cisco, Oct 2025

- Gained practical expertise in network topology design, routing protocols (OSPF, BGP), and troubleshooting using industry-standard Cisco tools.

LEADERSHIP & ACTIVITIES

Member - Club of Programmers (COPS), CBIT Kolar

Aug 2023 – Present

- Increased student engagement by 45%, measured by event participation growth from 180 to 200+ attendees, by orchestrating 2 technical workshops and 2 coding competitions.
- Improved mentee project completion rate to 80% measured across a 15-student cohort by providing technical mentorship in full-stack development, Data Structures & Algorithms (DSA), and Machine Learning (ML) fundamentals.
- Delivered successful CodeChallenge'25 event with 80+ participating teams by leading cross-functional coordination, including budget management, stakeholder engagement, and technical infrastructure setup.
- Enhanced knowledge transfer efficiency by 35% by facilitating peer-to-peer learning sessions on emerging technologies, including React.js, FastAPI, Docker containerization, and AI integration patterns.